



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

CHEMISTRY
PAPER 1

9189/1

JUNE 2018 SESSION

2 hours

Additional materials:

Answer paper

Data Booklet

Mathematical tables and/or electronic calculator

TIME: 2 hours

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number in the spaces provided on the answer paper/answer booklet.

Answer **six** questions.

Answer **two** questions from Section A, **one** question from Section B, **two** questions from Section C and **one** other question chosen from any section.

Write your answers on the separate answer paper provided.

If you use more than one sheet of paper, fasten the sheets together.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

You are reminded of the need for good English and clear presentation in your answers.

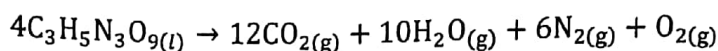
This question paper consists of 10 printed pages and 2 blank pages.

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Section A

Answer at least two questions from this section.

- 1 (a) (i) State the ideal gas equation.
- (ii) Nitroglycerine, $\text{C}_3\text{H}_5\text{N}_3\text{O}_9(l)$, is an explosive that decomposes according to the equation.

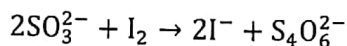


A 1.00 g sample of nitroglycerine was decomposed in a $1.20 \times 10^{-3} \text{ m}^3$ vessel at 800°C .

Calculate the pressure generated.

[5]

- (b) The active ingredient in most bleaching agents is sodium chlorate (I), NaOCl . A 10.00 cm^3 sample of the bleaching agent was diluted to 250.00 cm^3 . A 25.00 cm^3 portion of the dilute solution was mixed with an excess of potassium iodide, KI and dilute sulphuric acid. NaOCl , oxidises I^- to I_2 . The I_2 produced was titrated with 0.10 mol dm^{-3} solution of sodium thiosulphate and 21.00 cm^3 were needed to reach end point.



- (i) Name a suitable indicator for this titration.
- (ii) Write a balanced chemical equation for the oxidation of I^- by NaOCl .
- (iii) Calculate the
- number of moles of NaOCl in 25.00 cm^3 of the dilute bleach solution.
 - concentration of NaOCl in the original sample of the bleaching agent.

[7]

[Total:12]

- 2 (a) A chemical bond is formed between two atoms when their attractive forces outweigh their repulsive forces.

Identify the

- (i) attractive,
- (ii) repulsive forces.

[2]

- (b) Fig.2.1 shows the structure of the nitrate ion.

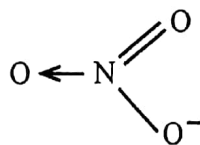


Fig.2.1

Describe the bonding in the nitrate ion.

[3]

- (c) (i) Define *hydrogen bonding*.
 (ii) State any **two** physical properties affected by hydrogen bonding.

[4]

- (d) (i) Arrange the given compounds in order of increasing solubility:

HI; MgO; MgCl₂;

- (ii) Explain the trend in (d)(i).

[3]

[Total:12]

- 3 Magnesium burns in air to give a compound, A and an oxide. Compound A consists of 72.25 % Mg and 27.75% N.

- (a) (i) Write a balanced equation for the formation of the oxide.
 (ii) Calculate the empirical formula of A.
 (iii) Write a balanced equation for the formation of A.
 (iv) Draw a dot and cross diagram to illustrate the bonding in A.
 (v) Explain why magnesium gives compound A, in addition to its oxide when burnt in air.

[7]

- (b) Table 3.1 shows enthalpy changes of three chemical processes.

Table 3.1

enthalpy change	energy value/ kJmol^{-1}
atomisation of magnesium	+ 148
total electron affinities for the change $\text{N}_{(\text{g})} \rightarrow \text{N}_{(\text{g})}^{3-}$	+ 2148
formation of compound A	-461

Calculate the lattice energy of Compound A.

[Use of data from the data booklet is recommended]

[5]

[Total:12]

Section B

Answer at least **one** question from this section.

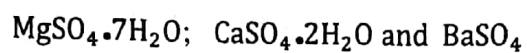
- 4 (a) Electronegativity values of halogens are shown in **Table 4.1**.

Table 4.1

halogen	F	Cl	Br	I
electronegativity	4.0	3.0	2.8	2.0

- (i) Define the term *electronegativity*.
- (ii) Suggest an explanation for the trend observed in the electronegativity values. [3]
- (b) Define the term *covalent atomic radius*. [1]
- (c) Iodide ions can reduce concentrated sulphuric acid to two different sulphur containing products.
- (i) State the observation made when the reaction occurs.
- (ii) Identify the **two** sulphur containing products.
- (iii) Give the oxidation state of sulphur in each product.
- (iv) Write balanced equations to show how each product is formed. [8]
- [Total:12]
- 5 (a) Carbon monoxide undergoes an exothermic, reversible disproportionation reaction under suitable conditions.
- (i) Write a balanced equation for this reaction.
- (ii) Use the reaction in (a)(i) to account for the relative stabilities of the +2 and +4 oxidation states of carbon. [3]
- (b) Write an equation for the reaction between carbon dioxide and sodium hydroxide. [1]

- (c) Some Group (II) sulphates are shown.



- (i) State any one use of

1. $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$,

2. $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$.

- (ii) Suggest an explanation for the variation in the number or moles of water of crystallisation in the sulphates. [4]

- (d) (i) Write an equation for the thermal decomposition of strontium nitrate.

- (ii) Calculate the volume of the gas produced when a 100 g sample of strontium nitrate is heated strongly. [4]

[Total;12]

7
Section C

Answer at least two question from this section.

- 6 (a) Phenylethene reacts with aqueous bromine at room temperature to form a di-bromo organic compound.

(i) Name the

1. type of reaction that takes place,
2. organic product of the reaction.

(ii) Describe the mechanism of the reaction.

[4]

- (b) Show how methylbenzene reacts with concentrated nitric acid to form two different products.

[2]

- (c) Fig. 6.1 shows how organic compounds B, C, and D are synthesized from compound A.

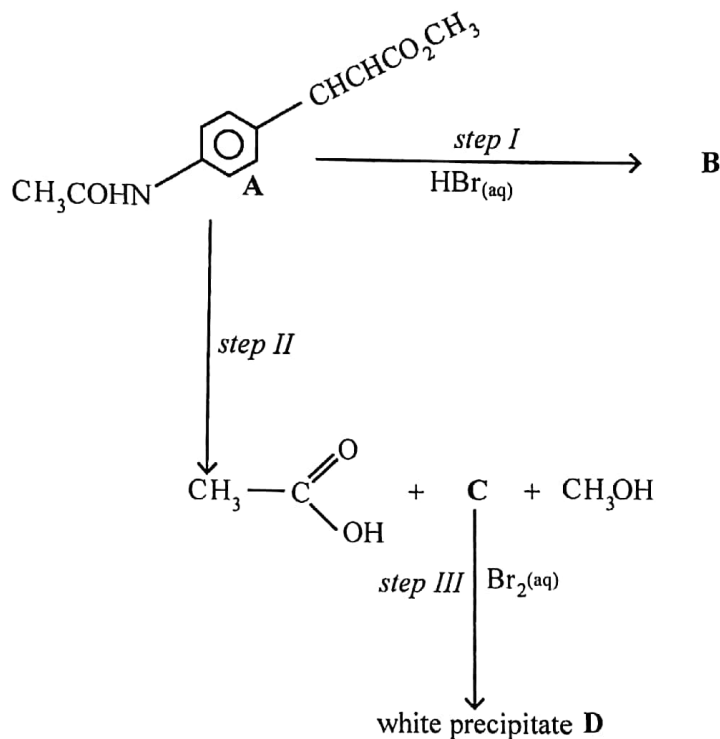


Fig. 6.1

(i) Give the structural formula of

1. **B,**
2. **C,**
3. **D.**

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[Turn over

- (ii) Suggest, with reasons, the acid-base nature of **B**.

[6]
[Total:12]

- 7 (a) The structural formula of compound **M** is shown in Fig.7.1.

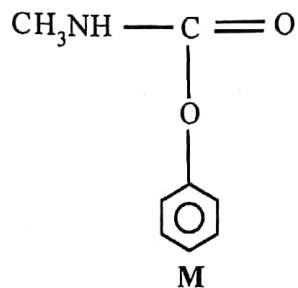


Fig.7.1

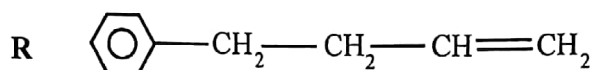
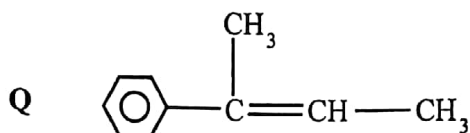
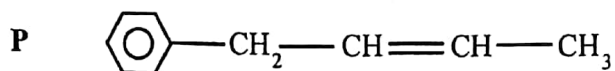
- (i) Name **two** functional groups present in **M**.
- (ii) State the reagents and conditions required for the hydrolysis of **M**.
- (iii) Write the structural formula of the product(s) formed when **M** is hydrolysed according to the hydrolysis conditions stated in (a) (ii).
- (b) Devise a stepwise reaction sequence for the conversion of benzene to a diazonium ion.

[6]

[6]

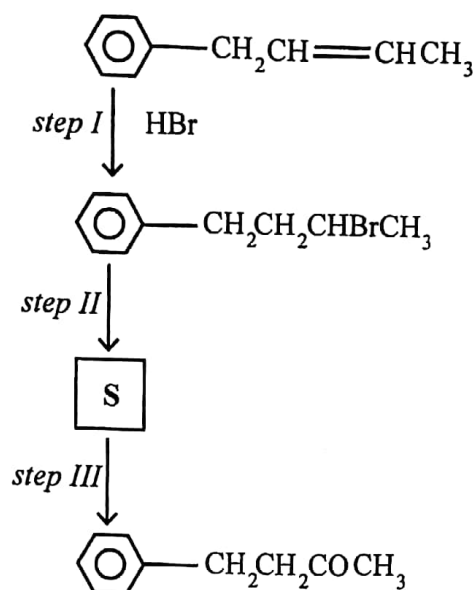
[Total:12]

8 Compounds **P**, **Q** and **R** are isomers.



- (a)
- (i) Write the molecular formula of these isomers.
 - (ii) Name the type of isomerism exhibited by **P** and **Q**.
 - (iii) Draw the structures of the pair of stereoisomers exhibited by **Q** and identify them. [4]

- (b) A reaction scheme for compound **P** is shown.



- (i) State the reagents and conditions for
1. *Step II*,
 2. *Step III*.
- (ii) Draw the structural formula of compound **S**.

[3]

- (c) The diacyl chloride, $\text{ClOCH}_2\text{CH}_2\text{OCl}$, reacts with ethane-1,2-diol to form a polymer.

- (i) Name the type of polymerisation that occurs.
- (ii) Draw the structural formula of
1. ethane-1,2-diol
 2. the product formed when one mole of the diacyl chloride reacts with two moles of ethane-1,2-diol.
- (iii) State any **one** physical property of the product in (c)(ii).2.
- (iv) Explain why ethane-1,2-diol is suitable for use as a coolant in automobile radiators.

[5]

[Total:12]